

Introduction to Probability and Statistics

Evans Gouno, University of South Brittany, France

evans.gouno@univ-ubs.fr

Syllabus

- Probability
- Inferential Statistics
 - Estimation
 - Nonparametric statistics
 - Bayesian Statistics

**AIMS**African Institute for
Mathematical Sciences
RWANDA

Lecture 1: Probability I

Introduction to Probability and Statistics

evans.gouno@univ-ubs.fr

- 1 Introduction
- 2 Random experiments
- 3 Events as sets
- 4 σ -field
- 5 Probability

Random experiments – Events

- Probability provides a mathematical framework for the study of random phenomena or **random experiments**.
- The goal is to measure the chance of **outcomes** that are produced by a random experiment.
- The collection of all the possible outcomes is called the **sample space** or **universe**.

It is usually denoted Ω .

- The **elementary outcomes** are denoted ω .
- An **event** is a subcollection of outcomes from Ω .

Example

- $\{\text{Rolling a die}\}$ is a random experiment.
- The universe for this experiment is:

$$\Omega = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}$$

It has 6 elements.

- $\{\text{The face containing three dots appears}\}$ is a possible elementary outcome.

$$\omega = \left\{ \begin{array}{|c|} \hline \bullet \bullet \\ \hline \end{array} \right\}.$$

- $\{\text{The result is odd}\}$ is an example of event .

$$A = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}$$

This event has 3 elements.

Events as sets

Exercise

- Random experiment: Toss a fair coin 10 times.

Describe Ω the sample space.

- Let A be the event '*At least one head occurs*'.

Give some examples of outcomes from A .

- Let T_j be the event that tails occurs on the j^{th} toss.

Describe T_j .

Events as sets

Solution

- The sample space Ω contains elements of the form $(c_1, c_2, \dots, c_{10})$ where $c_i = H$ or T , c_i being the result of the j^{th} toss.

$$\Omega = \{(c_1, c_2, \dots, c_{10}) \mid c_i = H \text{ or } T, i = 1, \dots, 10\}.$$

Example : $(T, T, H, T, H, H, T, T, T, T) \in \Omega$.

Remark that the number of elements in Ω is $2^{10} = 1024$.

- The event A contains all the element $(c_1, c_2, \dots, c_{10})$ of Ω such that at least one c_i is equal to H .

$$A = \{(c_1, c_2, \dots, c_{10}) \mid c_i = H \text{ and for } j \neq i, c_j = H \text{ or } T, j = 1, \dots, 10, \}.$$

Events as sets

Solution(cont.)

Some examples of outcomes from A : $(T, T, H, T, H, H, T, T, T, T)$,
 $(T, T, H, H, H, H, T, T, H, T)$, $(T, T, H, T, H, H, T, T, H, H)$.

Remark that only one element of Ω is not in A ; that is
 $(T, T, T, T, T, T, T, T, T, T)$.

- The event T_j contains all the element $(c_1, c_2, \dots, c_{10})$ of Ω such that $c_j = T$.

$$T_j = \{(c_1, \dots, c_{j-1}, T, c_{j+1}, \dots, c_{10}) \mid \text{for } i \neq j, c_i = H \text{ or } T, i = 1, \dots, 10.\}.$$

Some examples of outcomes from T_j :

$$\omega_1 = (T, T, H, T, \textcolor{red}{T}, H, T, T, T, T) \in T_5,$$

$$\omega_2 = (T, \textcolor{red}{T}, H, H, H, H, T, T, H, T) \in T_2,$$

$$\omega_3 = (T, T, H, T, H, H, \textcolor{red}{T}, T, H, H) \in T_7. \text{ Remark: } \omega_1, \omega_2 \text{ and } \omega_3 \text{ belong to } T_8.$$

Example

- $\{\text{Measuring some physical quantity}\}$, for example, temperature is a random experiment.
- In this case, Ω is $\mathbb{R} =]-\infty, +\infty[$, even if in practise temperature has a lower bound and an upper bound.
- Events are ranges of temperature as intervals in $\mathbb{R}$.
- For example, the event A that the temperature is larger than 12 but less than or equal to 18 is $A =]12, 18]$. The event that the temperature is below 0 is $] -\infty, 0[$.

Events as sets

Let A and B be two subsets of Ω .

The event ' A **or** B occur' is the subset $A \cup B$.

Definition

Union of two sets A and B :

$$A \cup B = \{\omega \in \Omega \mid \omega \in A \text{ or } \omega \in B\}.$$

The event ' A **and** B occur' is the subset $A \cap B$.

Definition

Intersection of two sets A and B :

$$A \cap B = \{\omega \in \Omega \mid \omega \in A \text{ and } \omega \in B\}.$$

Events as sets

Example:

Drawing a card from a deck, considering the events

A : 'the card is the king of diamonds'

B : 'the card is the ace of hearts'

→ obtaining A and B simultaneously is **impossible!**

An impossible event is represented by the empty set.

Definition

The empty set, denoted $\emptyset$, is the unique set such that for all $\omega \in \Omega$, $\omega \notin \emptyset$.

In other words, $\emptyset$ has no element.

Remark: $A \cap \emptyset = \emptyset$, $\forall A \subseteq \Omega$.

Events as sets

The event '*A does not occur*' means that the outcome belongs to the complement of A .

Definition

The complement of a set A is

$$\bar{A} = \{\omega \in \Omega \mid \omega \notin A\}.$$

Remarks:

- $A \cap \bar{A} = \emptyset$ and $A \cup \bar{A} = \Omega$.
- Performing an experiment leads necessarily to an outcome which is in Ω
 - ➡ Ω is the certain event.

Events as sets

The event '*A occurs, but not B*' is the set difference between *A* and *B*.

Definition

Difference between two sets A and B (set difference):

$$A \setminus B = A \cap \overline{B}.$$

Exercise: '*A and B do not occur simultaneously*'. Use the set theory jargon to describe this event?

Events as sets

The event '*A occurs, but not B*' is the set difference between A and B .

Definition

Difference between two sets A and B (set difference):

$$A \setminus B = A \cap \overline{B}.$$

Exercise: '*A and B do not occur simultaneously*'. Use the set theory jargon to describe this event?

$$(A \cup B) \setminus (A \cap B) = (A \cup B) \cap \overline{(A \cap B)}.$$

Events as sets

The event '*A occurs, but not B*' is the set difference between A and B .

Definition

Difference between two sets A and B (set difference):

$$A \setminus B = A \cap \overline{B}.$$

Exercise: '*A and B do not occur simultaneously*'. Use the set theory jargon to describe this event?

$$(A \cup B) \setminus (A \cap B) = (A \cup B) \cap \overline{(A \cap B)}.$$

'*A and B cannot occur simultaneously*' means that A and B are disjoint.

Definition

A and B are disjoint iff $A \cap B = \emptyset$.

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
- **Associative laws** :
 - $A \cap (B \cap C) = (A \cap B) \cap C$,

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
- **Associative laws** :
 - $A \cap (B \cap C) = (A \cap B) \cap C$,
 - $A \cup (B \cup C) = (A \cup B) \cup C$.

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
- **Associative laws** :
 - $A \cap (B \cap C) = (A \cap B) \cap C$,
 - $A \cup (B \cup C) = (A \cup B) \cup C$.
- **Distributive laws** :
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
- **Associative laws** :
 - $A \cap (B \cap C) = (A \cap B) \cap C$,
 - $A \cup (B \cup C) = (A \cup B) \cup C$.
- **Distributive laws** :
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Events as sets

Let A, B, C be 3 subsets of Ω , we have:

- **Idempotency laws** : $A \cap A = A$ and $A \cup A = A$.
- **Commutative laws** : $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
- **Associative laws** :
 - $A \cap (B \cap C) = (A \cap B) \cap C$,
 - $A \cup (B \cup C) = (A \cup B) \cup C$.
- **Distributive laws** :
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Events as sets

- $A \cup \emptyset =$

Events as sets

- $A \cup \emptyset = A$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega =$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$  Ω acts as an **identity element** for the intersection.

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$  Ω acts as an **identity element** for the intersection.
- $\overline{\overline{A}} =$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$  Ω acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.

Events as sets

- $A \cup \emptyset = A$ $\rightarrow \emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$ $\rightarrow \Omega$ acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$

Events as sets

- $A \cup \emptyset = A$ $\rightarrow \emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$ $\rightarrow \Omega$ acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$  Ω acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$
 $B \subseteq A \iff A \cup B = A$

Events as sets

- $A \cup \emptyset = A$  $\emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$  Ω acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$
 $B \subseteq A \iff A \cup B = A$.

Events as sets

- $A \cup \emptyset = A$ $\rightarrow \emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$ $\rightarrow \Omega$ acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$
 $B \subseteq A \iff A \cup B = A$.
- De Morgan's Laws : $\overline{A \cup B} =$

Events as sets

- $A \cup \emptyset = A$ $\rightarrow \emptyset$ acts as an **identity element** for the union.
- $A \cap \Omega = A$ $\rightarrow \Omega$ acts as an **identity element** for the intersection.
- $\overline{\overline{A}} = A$.
- $B \subseteq A \iff A \cap B = B$
 $B \subseteq A \iff A \cup B = A$.
- De Morgan's Laws : $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Proof?

Events as sets

Proof of De Morgan's laws:

- For any ω in Ω , if $\omega \in \overline{A \cup B}$, $\omega \notin A \cup B$ which means that ω is neither in A nor in B . Therefore $\omega \in \overline{A}$ **and** $\omega \in \overline{B}$.
Thus $\omega \in \overline{A} \cap \overline{B}$. Then $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$.
- Conversely, if $\omega \in \overline{A} \cap \overline{B}$, $\omega \notin A$ **and** $\omega \notin B$. Then $\omega \notin A \cup B$.
Therefore $\omega \in \overline{A \cup B}$. Thus $\overline{A} \cap \overline{B} \subseteq \overline{A \cup B}$.

Conclusion:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}.$$

Events as sets

Proof of De Morgan's laws:

- For any ω in Ω , if $\omega \in \overline{A \cup B}$, $\omega \notin A \cup B$ which means that ω is neither in A nor in B . Therefore $\omega \in \overline{A}$ **and** $\omega \in \overline{B}$.
Thus $\omega \in \overline{A} \cap \overline{B}$. Then $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$.
- Conversely, if $\omega \in \overline{A} \cap \overline{B}$, $\omega \notin A$ **and** $\omega \notin B$. Then $\omega \notin A \cup B$.
Therefore $\omega \in \overline{A \cup B}$. Thus $\overline{A} \cap \overline{B} \subseteq \overline{A \cup B}$.

Conclusion:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}.$$

Applying this result to $\overline{A}$ and $\overline{B}$, we have:

$$\overline{\overline{A \cup B}} = \overline{\overline{A} \cap \overline{B}} \iff \overline{\overline{A \cup B}} = A \cap B.$$

Taking the complement of each side of the last equality we obtain:

$$\overline{A \cup B} = \overline{A \cap B}.$$

Events as Sets

Events	Set
Universe (sample space)	Ω
Outcome (elementary event)	$\{\omega\}$
Impossible event (null event)	$\emptyset$
Both A and B occur	$A \cap B$
Either A or B or both occur	$A \cup B$
A occurs but not B	$A \setminus B$
A and B are incompatible	$A \cap B = \emptyset$
A produces B ($A \implies B$)	$A \subset B$

Events as sets

Exercise

Let A, B, C be 3 events.

Consider $E_1 = A \cap \overline{B} \cap \overline{C}$ and $E_2 = A \cap (B \cup C)$.

- ① Show that E_1 and E_2 are disjoint.
- ② What is $E_1 \cup E_2$?

Events as sets

Exercise

Let A, B, C be 3 events.

Consider $E_1 = A \cap \overline{B} \cap \overline{C}$ and $E_2 = A \cap (B \cup C)$.

① Show that E_1 and E_2 are disjoint.

② What is $E_1 \cup E_2$?

Solution

① E_1 and E_2 are disjoint is equivalent to $E_1 \cap E_2 = \emptyset$

$$\begin{aligned} E_1 \cap E_2 &= (A \cap \overline{B} \cap \overline{C}) \cap (A \cap (B \cup C)) = A \cap [(\overline{B} \cap \overline{C}) \cap (B \cup C)] \\ &= A \cap [\overline{B \cup C} \cap (B \cup C)] = A \cap \emptyset = \emptyset. \end{aligned}$$

Events as sets

Exercise

Let A, B, C be 3 events.

Consider $E_1 = A \cap \overline{B} \cap \overline{C}$ and $E_2 = A \cap (B \cup C)$.

① Show that E_1 and E_2 are disjoint.

② What is $E_1 \cup E_2$?

Solution

① E_1 and E_2 are disjoint is equivalent to $E_1 \cap E_2 = \emptyset$

$$\begin{aligned} E_1 \cap E_2 &= (A \cap \overline{B} \cap \overline{C}) \cap (A \cap (B \cup C)) = A \cap [(\overline{B} \cap \overline{C}) \cap (B \cup C)] \\ &= A \cap [\overline{B \cup C} \cap (B \cup C)] = A \cap \emptyset = \emptyset. \end{aligned}$$

②

$$\begin{aligned} E_1 \cup E_2 &= (A \cap \overline{B} \cap \overline{C}) \cup (A \cap (B \cup C)) = A \cap [(\overline{B} \cap \overline{C}) \cup (B \cup C)] \\ &= A \cap [(\overline{B \cup C}) \cup (B \cup C)] = A \cap \Omega = A. \end{aligned}$$

σ -field

Events are subsets of Ω but in some cases, all subsets of Ω are not events.

We will consider a subcollection $\mathcal{F}$ of the set of all subsets of Ω .

Expected properties for $\mathcal{F}$

- if $A, B \in \mathcal{F}$ then $A \cup B \in \mathcal{F}$,
- if $A \in \mathcal{F}$ then $\bar{A} \in \mathcal{F}$,
- the empty set $\emptyset$ belongs to $\mathcal{F}$.

Any collection $\mathcal{F}$ of subsets of Ω which satisfies these three conditions is called a **field** or an **algebra**.

Consequently $\Rightarrow$ If $A_1, A_2, \dots, A_n \in \mathcal{F}$ then $\bigcup_{i=1}^n A_i \in \mathcal{F}$.

σ -field

In order to be able to discuss probability we need more; we require that the collection of events be closed under the operation of taking any countable unions of elements from $\mathcal{F}$.

➡ Then the collection becomes a σ -field.

Definition

A collection $\mathcal{F}$ of subsets of Ω is called a **σ -field** if it satisfies the following conditions:

- $\emptyset \in \mathcal{F}$,
- if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
- if $A \in \mathcal{F}$ then $\bar{A} \in \mathcal{F}$.

σ -field**Exercise**

Let A be a subset of Ω .

Prove that $\mathcal{F} = \{\emptyset, A, \bar{A}, \Omega\}$ is a σ -field.

Solution

The conditions are easily checked

- The empty set is in $\mathcal{F}$.
- We have for unions:

$$\begin{aligned}\emptyset \cup A &= A \in \mathcal{F}, & A \cup \bar{A} &= \Omega \in \mathcal{F}, & \emptyset \cup A \cup \bar{A} &= \Omega \in \mathcal{F}, \\ \emptyset \cup \bar{A} &= \bar{A} \in \mathcal{F}, & A \cup \Omega &= \Omega \in \mathcal{F}, & \emptyset \cup A \cup \Omega &= \Omega \in \mathcal{F}, \\ \emptyset \cup \Omega &= \Omega \in \mathcal{F}, & \bar{A} \cup \Omega &= \Omega \in \mathcal{F}, & \emptyset \cup \bar{A} \cup \Omega &= \Omega \in \mathcal{F}, \\ A \cup \bar{A} \cup \Omega &= \Omega \in \mathcal{F}, & \emptyset \cup A \cup \bar{A} \cup \Omega &= \Omega \in \mathcal{F}.\end{aligned}$$

- Complement:

$$\bar{\emptyset} = \Omega \in \mathcal{F}, \quad \bar{A} \in \mathcal{F}, \quad \overline{\bar{A}} = A \in \mathcal{F}, \quad \bar{\Omega} = \emptyset \in \mathcal{F}.$$

Probability on finite sample space

- Our goal is now to measure the chance for an event to occur.
- To proceed we construct a function P that assigns a unique number to each event.
- A natural way:

$$P(A) = \frac{\#A}{\#\Omega}.$$

The function P is such that $0 \leq P(A) \leq 1$, $P(\Omega) = 1$ and $P(\emptyset) = 0$.

n.b. here Ω is assumed to be finite and countable.

Example

- Random experiment: *Rolling a fair die.*
- The universe for this experiment is:

$$\Omega = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}.$$

➡ $\#\Omega = 6.$

- $A = \{ \text{The result is odd} \} = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}.$

➡ $\#A = 3.$

- Probability for the result to be odd:

Example

- Random experiment: *Rolling a fair die.*
- The universe for this experiment is:

$$\Omega = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}.$$

➡ $\#\Omega = 6.$

- $A = \{ \text{The result is odd} \} = \left\{ \begin{array}{|c|} \hline \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \\ \hline \end{array}, \begin{array}{|c|} \hline \bullet \bullet \bullet \bullet \bullet \\ \hline \end{array} \right\}.$

➡ $\#A = 3.$

- Probability for the result to be odd:

$$P(A) = \frac{\#A}{\#\Omega} = \frac{3}{6} = \frac{1}{2}.$$

Probability on finite sample spaces

Exercise – A die is rolled twice. Display the sample space for this experiment by listing its elements in full detail. On a diagram, circle the following events:

A : faces are equal.

B : sum of faces is ≤ 10 .

C : faces sum to 7.

Compute $P(A)$, $P(B)$ and $P(C)$.